

# Analytic Expressions for Periods and Quasi-Periods at Weight 12 via the Diamantis–Rolen Finite-Endpoint Cocycle

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## Abstract

We record explicit analytic formulae for the periods  $\omega^\pm$  of the modular discriminant  $\Delta \in S_{12}$  and the quasi-periods  $\eta^\pm$  of the weakly holomorphic form  $\hat{\Delta} \in S_{12}^!$ , both extracted from the same Diamantis–Rolen (DR) finite-endpoint cocycle construction. For  $\Delta$  the formulae reduce to ratios of standard critical  $L$ -values; for  $\hat{\Delta}$  they define the correct DR-regularised analogues, which differ from BFK’s non-holomorphic regularisation and are the unique values landing in the cuspidal cohomology  $H_{\text{cusp}}^1(\text{SL}_2(\mathbb{Z}); V_{10})$ . All numerical values are verified to  $\sim 20$ – $32$  significant digits in the companion notebook `period_extraction_DR_general.ipynb`.

## 1 Setup

Let  $\Gamma = \text{SL}_2(\mathbb{Z})$ ,  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ , weight  $k = 12$ ,  $n = k - 2 = 10$ .

The polynomial space  $V_{10} = \bigoplus_{i+j=10} \mathbb{Q}X^iY^j$  carries the right  $\Gamma$ -action  $(X, Y)|_\gamma = (aX + bY, cX + dY)$ . Brown’s basis [1] for the cuspidal cohomology  $H_{\text{cusp}}^1(\Gamma; V_{10})$  has  $S$ -values

$$P_S^+ = \frac{36}{691}(Y^{10} - X^{10}) + X^2Y^2(X^2 - Y^2)^3, \quad (1)$$

$$P_S^- = 4X^9Y - 25X^7Y^3 + 42X^5Y^5 - 25X^3Y^7 + 4XY^9. \quad (2)$$

We write  $[P]_{X^{10-j}Y^j}$  for the coefficient of  $X^{10-j}Y^j$  in a polynomial  $P \in V_{10}$ .

## 2 The DR cocycle and period extraction

### 2.1 Cocycle values as polynomials in moments of $f$

Fix  $\tau_0 \in \mathbb{H}$ . For any  $f$  of weight  $k = 12$  on  $\Gamma$ , define the **moment integrals**

$$N_j(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau, \quad M_j(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^j d\tau, \quad j = 0, 1, \dots, 10. \quad (3)$$

Both are well-defined finite-contour integrals of  $f$  over straight line segments in  $\mathbb{H}$  (the  $S$ -segment and the  $T$ -segment respectively); for  $f \in S_{12}^!$  no regularisation is required because the contours stay in  $\mathbb{H}$ . Expanding  $(X - \tau Y)^{10}$  binomially in (4),(5) below, the cocycle values at  $S$  and  $T$  are

explicit polynomials with the moments as coefficients:

$$C_S^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} N_j X^{10-j} Y^j, \quad (4)$$

$$C_T^f(X, Y) = (2\pi i)^{11} \sum_{j=0}^{10} (-1)^j \binom{10}{j} M_j X^{10-j} Y^j. \quad (5)$$

Cuspidality of the cohomology class  $[C^f] \in H^1(\Gamma; V_{10})$  is equivalent to

$$[C_T^f]_{X^{10}} = (2\pi i)^{11} M_0(f, \tau_0) = 0, \quad (6)$$

which holds whenever the constant Fourier coefficient  $c_f(0)$  vanishes (since  $M_0 = \int_{\tau_0-1}^{\tau_0} f = c_f(0)$  by periodicity). For both  $\Delta$  and  $\hat{\Delta}$  this is automatic.

## 2.2 Moments as explicit BFK-style sums over $q$ -coefficients

Substituting the Fourier expansion  $f(\tau) = \sum_n a_f(n) e^{2\pi i n \tau}$  into (3) and integrating term-by-term gives explicit closed forms. Set  $q_0 := e^{2\pi i \tau_0}$  and  $\tilde{q}_0 := e^{-2\pi i / \tau_0}$ ; both satisfy  $|q_0|, |\tilde{q}_0| < 1$  for any  $\tau_0 \in \mathbb{H}$ . Repeated integration by parts on  $\int_a^b e^{2\pi i n \tau} \tau^j d\tau$  (for  $n \neq 0$ ) yields the elementary identity

$$\int_a^b e^{2\pi i n \tau} \tau^j d\tau = \sum_{k=0}^j \frac{(-1)^k j!}{(j-k)! (2\pi i n)^{k+1}} \left[ e^{2\pi i n b} b^{j-k} - e^{2\pi i n a} a^{j-k} \right]. \quad (7)$$

Equivalently, in terms of upper incomplete gamma functions  $\Gamma(s, x) := \int_x^\infty t^{s-1} e^{-t} dt$  (which reduce to the elementary form  $\Gamma(j+1, x) = j! e^{-x} \sum_{k=0}^j x^k / k!$  at integer first argument),

$$\int_a^b e^{2\pi i n \tau} \tau^j d\tau = \frac{i^{j-1}}{(2\pi n)^{j+1}} \left[ \Gamma(j+1, -2\pi i n b) - \Gamma(j+1, -2\pi i n a) \right]. \quad (8)$$

Applied to the  $T$ -segment  $[\tau_0 - 1, \tau_0]$ , using  $e^{2\pi i n(\tau_0-1)} = q_0^n$  for integer  $n$ :

$$\boxed{M_j(f, \tau_0) = a_f(0) \frac{\tau_0^{j+1} - (\tau_0 - 1)^{j+1}}{j+1} + \sum_{n \neq 0} a_f(n) q_0^n \mathcal{K}_j^T(n, \tau_0),} \quad (9)$$

$$\mathcal{K}_j^T(n, \tau_0) := \sum_{k=0}^j \frac{(-1)^k j!}{(j-k)! (2\pi i n)^{k+1}} \left[ \tau_0^{j-k} - (\tau_0 - 1)^{j-k} \right]. \quad (10)$$

Applied to the  $S$ -segment  $[-1/\tau_0, \tau_0]$ :

$$\boxed{N_j(f, \tau_0) = a_f(0) \frac{\tau_0^{j+1} - (-1/\tau_0)^{j+1}}{j+1} + \sum_{n \neq 0} a_f(n) \mathcal{K}_j^S(n, \tau_0),} \quad (11)$$

$$\mathcal{K}_j^S(n, \tau_0) := \sum_{k=0}^j \frac{(-1)^k j!}{(j-k)! (2\pi i n)^{k+1}} \left[ q_0^n \tau_0^{j-k} - \tilde{q}_0^n (-1/\tau_0)^{j-k} \right]. \quad (12)$$

**Convergence.** For  $f \in S_{12}$  only  $n \geq 1$  contributes, with  $a_f(n) = O(n^{11/2+\epsilon})$  and  $|q_0|^n, |\tilde{q}_0|^n$  decaying as  $e^{-2\pi n \operatorname{Im}(\tau_0)}$  and  $e^{2\pi n \operatorname{Im}(-1/\tau_0)}$  respectively — both exponential, both contracting. For  $f = \hat{\Delta} \in S_{12}^!$  the only modifications are: (i) the sum runs from  $n = -1$ , contributing one finite explicit term  $q_0^{-1} \mathcal{K}_j^T(-1, \tau_0)$  to  $M_j$  (and analogously to  $N_j$ ); (ii) for  $n \geq 2$  the  $q$ -coefficients grow as  $a_f(n) \sim e^{4\pi\sqrt{n}}$ , but  $|q_0|^n$  still decays super-exponentially in  $n$ , so the series *converges with no regularisation needed* for any  $\tau_0$  in the interior of  $\mathbb{H}$  — the basepoint, not analytic continuation, is what tames the cusp pole. This is the structural reason DR succeeds where BFK's single-parameter regulator stumbles.

**The full chain.** Putting it all together: substituting (9) into the moments appearing in (16) expresses  $P(X, Y)$  as an explicit double sum over  $\{a_f(n)\}$  and monomials  $\{X^a Y^{10-a}\}$ , with weights  $B_k j! \binom{10}{j} (2\pi i)^{11} q_0^n \mathcal{K}_j^T(n, \tau_0)$ . Substituting (11) into (4) does the same for  $C_S^f$ . The modified cocycle (18) then reads

$$\boxed{C_S^f(X, Y) = a_f(0) \Phi_0(X, Y; \tau_0) + \sum_{n \neq 0} a_f(n) \Phi_n(X, Y; \tau_0),} \quad (13)$$

where each  $\Phi_n(X, Y; \tau_0) \in V_{10} \otimes \mathbb{C}$  is an *explicit polynomial* in  $X, Y$  with coefficients given by (10), (12), the Bernoulli weights of (15), and the  $S$ -image substitution  $X \leftrightarrow Y, Y \leftrightarrow -X$ . In particular, the periods  $\omega^\pm(f)$  are extracted via (21) as

$$\omega^\pm(f) = a_f(0) \alpha_0^\pm(\tau_0) + \sum_{n \neq 0} a_f(n) \alpha_n^\pm(\tau_0), \quad (14)$$

with each  $\alpha_n^\pm(\tau_0) \in \mathbb{C}$  an explicit elementary function of  $\tau_0, q_0^n, \tilde{q}_0^n$ , and  $1/(2\pi i n)^{k+1}$ . This is the fully explicit BFK-style sum that determines  $\omega^\pm(\Delta)$  and  $\eta^\pm(\hat{\Delta})$  uniformly in the same closed form.

### 2.3 Closed-form coboundary via Bernoulli numbers

Viewing the slash action of  $T$  on polynomials as the translation operator  $T = e^{Y\partial_X}$ , the equation  $P|_T - P = C_T^f$  is exactly  $(e^{Y\partial_X} - 1)P = C_T^f$ . The Bernoulli generating identity  $z/(e^z - 1) = \sum_{k \geq 0} B_k z^k/k!$  ( $B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_3 = 0, B_4 = -\frac{1}{30}, \dots$ , with  $B_{2\ell+1} = 0$  for  $\ell \geq 1$ ) gives the formal inverse  $(e^{Y\partial_X} - 1)^{-1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}$ , which terminates on a polynomial of finite  $X$ -degree. Reading off the  $k = 0$  term as the antiderivative  $\partial_X^{-1} = \int_0^X \cdot dt$ , we obtain the **explicit closed form**

$$\boxed{P(X, Y) = \frac{1}{Y} \int_0^X C_T^f(t, Y) dt + \sum_{k=1}^{10} \frac{B_k}{k!} Y^{k-1} \frac{\partial^{k-1} C_T^f}{\partial X^{k-1}}(X, Y).} \quad (15)$$

Substituting (5) and integrating/differentiating monomial-by-monomial:

$$P(X, Y) = (2\pi i)^{11} \sum_{j=1}^{10} (-1)^j \binom{10}{j} M_j \left[ \frac{X^{11-j} Y^{j-1}}{11-j} + \sum_{k=1}^{11-j} \frac{B_k (10-j)!}{k! (11-j-k)!} X^{11-j-k} Y^{j+k-1} \right]. \quad (16)$$

(The  $j = 0$  term drops by cuspidality (6).) Writing  $p_j$  for the coefficient of  $X^{10-j} Y^j$  in  $P$ , the first three values are

$$p_0 = -(2\pi i)^{11} M_1, \quad p_1 = 5 (2\pi i)^{11} (M_1 + M_2), \quad p_2 = -(2\pi i)^{11} \left( \frac{15}{2} M_1 + \frac{45}{2} M_2 + 15 M_3 \right), \quad (17)$$

with the pattern continuing through  $p_9$  via (16); the  $p_{10}$  coefficient is the only direction unconstrained by  $P|_T - P = C_T^f$  (namely  $Y^{10}$ , which is  $T$ -fixed) and is harmless: see below. Agreement of (15) with the upper-triangular recursion solver used in the main pipeline is verified to  $\sim 10^{-42}$  relative residual at 45-digit working precision in [5], on every  $p_m$  for  $m = 0, \dots, 9$ .

## 2.4 The modified cocycle value as an explicit polynomial

With  $P$  given by (16) and  $P|_S(X, Y) = P(Y, -X)$ , define

$$C_S'^f(X, Y) := C_S^f(X, Y) - [P(Y, -X) - P(X, Y)]. \quad (18)$$

Both pieces are now explicit polynomials in  $V_{10} \otimes \mathbb{C}$  with coefficients that are explicit linear combinations of the moments  $\{N_j, M_j\}_{j=0}^{10}$  (rational/Bernoulli weights, overall factor  $(2\pi i)^{11}$ ). Concretely,

$$[C_S'^f]_{X^{10-j}Y^j} = (2\pi i)^{11}(-1)^j \binom{10}{j} N_j - [P(Y, -X) - P(X, Y)]_{X^{10-j}Y^j}, \quad (19)$$

where the bracket on the right is the explicit polynomial difference of (16) and its  $S$ -image.

By construction  $C_T'^f = 0$ , so  $C_S'^f$  is the  $S$ -value of a cuspidal 1-cocycle and decomposes uniquely in Brown's basis as

$$C_S'^f = \omega^+(f) P_S^+ + \omega^-(f) P_S^- + c(X^{10} - Y^{10}), \quad (20)$$

with  $c \in \mathbb{C}$  depending on the choice of  $p_{10}$  (the unconstrained  $Y^{10}$  direction). Replacing  $P \rightarrow P + c' Y^{10}$  shifts  $C_S'^f$  by  $c'(X^{10} - Y^{10})$ , which is itself a coboundary in  $V_{10}$  — thus the cohomology class is unchanged, and so are  $\omega^\pm(f)$ .

## 2.5 Period extraction

Reading off interior monomials (those for which  $X^{10} - Y^{10}$  has zero coefficient) gives  $\omega^\pm$  unambiguously:

$$\omega^-(f) = \frac{[C_S'^f]_{X^9Y}}{4}, \quad \omega^+(f) = [C_S'^f]_{X^8Y^2}. \quad (21)$$

Internal consistency is verified by repeating from all five odd (resp. four even) interior monomials.

# 3 Periods as DR-regularised $L$ -values

## 3.1 Definition of $L_{\text{reg}}(f, s)$ entirely from the DR cocycle

For any weight-12 form  $f$  on  $\Gamma$  (holomorphic or weakly holomorphic) and any  $\tau_0 \in \mathbb{H}$ , **define**

$$L_{\text{reg}}(f, j+1) := \frac{[C_S'^f(\tau_0)]_{X^{10-j}Y^j}}{(2\pi)^{10-j}(-i)^j \binom{10}{j} j!}, \quad j = 0, 1, \dots, 10. \quad (22)$$

The right-hand side is constructed entirely from the DR cocycle:  $C_S'^f(\tau_0) = C_S^f - (P|_S - P)$  is built from the finite-contour moments (3) via (4), (5), and the closed-form Bernoulli formula (15); equivalently, it is the explicit  $q$ -coefficient sum (13). No integration along  $0 \rightarrow i\infty$  enters anywhere.<sup>1</sup> The right side of (22) is a priori a function of  $\tau_0$ , but is in fact  $\tau_0$ -independent (Section 4).

<sup>1</sup>For  $f \in S_{12}$ , the standard Mellin identity  $\int_0^{i\infty} f(\tau) \tau^j d\tau = i^{j+1} j! / (2\pi)^{j+1} \cdot L(f, j+1)$  — invoked only as a parenthetical check — shows  $L_{\text{reg}}(f, s) = L(f, s)$  for the standard analytically continued  $L$ -function. For  $f \in S_{12}^!$ , (22) defines the correct cohomological  $L$ -value, finite at any  $\tau_0 \in \mathbb{H}$ .

### 3.2 Explicit period formulas with all numbers plugged in

Substituting (22) into the extraction formula (21) and using the explicit Brown-basis values  $[P_S^-]_{X^9Y, X^7Y^3, X^5Y^5, X^3Y^7}$  (4, -25, 42, -25, 4),  $[P_S^+]_{X^8Y^2, X^6Y^4, X^4Y^6, X^2Y^8} = (1, -3, 3, -1)$  from Brown [1] §8.2 (and Section 1 above), the periods read explicitly as:

$$\begin{aligned}\omega^-(f) &= -\frac{5}{2}i(2\pi)^9 L_{\text{reg}}(f, 2) = -\frac{144}{5}i(2\pi)^7 L_{\text{reg}}(f, 4) = -720i(2\pi)^5 L_{\text{reg}}(f, 6) \\ &= -24192i(2\pi)^3 L_{\text{reg}}(f, 8) = -1814400i\pi L_{\text{reg}}(f, 10),\end{aligned}\tag{23}$$

$$\begin{aligned}\omega^+(f) &= -90(2\pi)^8 L_{\text{reg}}(f, 3) = -1680(2\pi)^6 L_{\text{reg}}(f, 5) = -50400(2\pi)^4 L_{\text{reg}}(f, 7) \\ &= -1814400(2\pi)^2 L_{\text{reg}}(f, 9).\end{aligned}\tag{24}$$

The five (resp. four) equalities encode the lockstep rationality among critical  $L$ -values; for  $f \in S_{12}$  they recover the classical relations.

**For  $f = \Delta$ :** (23), (24) are the classical period  $\omega^\pm(\Delta) = \omega^\pm$  formulas, with  $L_{\text{reg}}(\Delta, s) = L(\Delta, s)$  the standard critical  $L$ -value. The notebook's  $L$ -values for  $\Delta$  at  $\tau_0 = 0.3 + 1.2i$  are:

| $s$ | $L_{\text{reg}}(\Delta, s)$ | feeds into    | recovered period                     |
|-----|-----------------------------|---------------|--------------------------------------|
| 2   | 0.14637454209126598...      | (23), $j = 1$ | $\omega^- = -5585015.379310401... i$ |
| 3   | 0.31524156588099308...      | (24), $j = 2$ | $\omega^+ = -68916772.80959519...$   |
| 4   | 0.50161764751090221...      | (23), $j = 3$ | $\omega^- = -5585015.379310401... i$ |
| 5   | 0.66670918843400364...      | (24), $j = 4$ | $\omega^+ = -68916772.80959519...$   |
| 6   | 0.79212283864603056...      | (23), $j = 5$ | $\omega^- = -5585015.379310401... i$ |
| 7   | 0.87735412538866091...      | (24), $j = 6$ | $\omega^+ = -68916772.80959519...$   |
| 8   | 0.93070703029812609...      | (23), $j = 7$ | $\omega^- = -5585015.379310401... i$ |
| 9   | 0.96212645969442586...      | (24), $j = 8$ | $\omega^+ = -68916772.80959519...$   |
| 10  | 0.97980908825122051...      | (23), $j = 9$ | $\omega^- = -5585015.379310401... i$ |

All nine rows agree to relative error  $< 6 \times 10^{-42}$  at 45-digit working precision, confirming the lockstep relations in (23) and (24) to numerical precision [6].

**For  $f = \hat{\Delta}$ :** (23), (24) define the quasi-periods  $\eta^\pm(\hat{\Delta})$  via  $L_{\text{reg}}(\hat{\Delta}, s)$ . The  $L_{\text{reg}}$  values cannot be recovered from BFK's  $L_{\text{BKF}}^*(\hat{\Delta}, s)$ : the latter satisfies  $(1 + S) = 0$  but fails  $(1 + U + U^2) = 0$ , while  $L_{\text{reg}}$  comes from a genuine cocycle (Section 4).

## 4 Basepoint independence as an explicit coboundary

The right side of (22) appears to depend on  $\tau_0$ ; it does not. The reason is an explicit coboundary identity that propagates through the whole construction. Define the basepoint-shift potential

$$h(X, Y; \tau_0, \tau_1) := (2\pi i)^{11} \int_{\tau_0}^{\tau_1} f(\tau) (X - \tau Y)^{10} d\tau \in V_{10} \otimes \mathbb{C}.\tag{25}$$

A short computation using the modular property  $\int_{\gamma^{-1}\tau_a}^{\gamma^{-1}\tau_b} f(\tau) (X - \tau Y)^{10} d\tau = h(X, Y; \tau_a, \tau_b)|_\gamma$  (which follows from the weight-12 transformation of  $f(\tau) d\tau$  together with the slash-action on  $(X - \tau Y)^{10}$ ) gives

$$\boxed{C_\gamma^f(\tau_1) - C_\gamma^f(\tau_0) = -\left[h(\tau_0, \tau_1)|_\gamma - h(\tau_0, \tau_1)\right], \quad \forall \gamma \in \Gamma.}\tag{26}$$

The right side is manifestly a coboundary in  $V_{10} \otimes \mathbb{C}$ , so the cohomology class  $[C_\bullet^f] \in H^1(\Gamma; V_{10} \otimes \mathbb{C})$  is  $\tau_0$ -independent. The cuspidal coboundary modification  $C_S'^f$  inherits this invariance, hence  $\omega^\pm(f)$  — and therefore  $L_{\text{reg}}(f, s)$  — are  $\tau_0$ -independent.

#### 4.1 Analytic check from the explicit formulas

For an explicit nerd-friendly check: substitute (9), (11) into the master formula (13). Each  $q$ -coefficient  $a_f(n)$  multiplies a  $\tau_0$ -dependent function  $\Phi_n(X, Y; \tau_0)$  that involves  $q_0^n, \tilde{q}_0^n, \tau_0^j, (\tau_0 - 1)^j, (-1/\tau_0)^j$ , and Bernoulli rationals. The same  $q$ -coefficient  $a_f(n)$  also enters  $h(X, Y; \tau_0, \tau_1)$  via term-by-term integration of  $\int_{\tau_0}^{\tau_1} f(\tau)(X - \tau Y)^{10} d\tau$ , which by (7) produces only boundary evaluations  $[e^{2\pi i n \tau} P_{j,n}(\tau)]_{\tau_0}^{\tau_1}$  for explicit polynomials  $P_{j,n}$ . Hence (26) reduces to a polynomial identity satisfied term-by-term in  $a_f(n)$ , with the  $\tau_0$ -dependence matching exactly between the two sides.

**Meromorphic generalisation.** For a meromorphic modular form  $f$  with poles inside  $\mathbb{H}$  (e.g.  $1/E_6$ , which has a simple pole at  $\tau = i$  and modular images thereof), the same construction goes through with one modification: the contour from  $\gamma^{-1}\tau_0$  to  $\tau_0$  may need to be deformed around interior poles of  $f$ , and the basepoint identity (26) acquires an extra residue term whenever a basepoint shift moves a pole across the contour. That residue is itself an explicit polynomial in  $V_{10}$ , hence still a coboundary, leaving the cohomology class — and the periods — invariant. This is the natural extension of the present formalism to fully meromorphic modular forms with poles both at the cusp and in the interior of  $\mathbb{H}$ .

### 5 Comparison with BFK's regularisation

The DR cocycle (18) is finite for both  $\Delta$  and  $\hat{\Delta}$  at any  $\tau_0 \in \mathbb{H}$ : every contour is a finite line segment, every  $q$ -mode integrates to a closed-form expression (7), and the notebook computes all coefficients to  $\sim 32$  digits. No regularisation enters.

BFK's construction is also convergent. It defines finite regularised  $L$ -values  $L_{\text{BFK}}^*(\hat{\Delta}, s)$  via a non-holomorphic exponential factor  $e^{2\pi\nu \text{Im}(\tau)}$  and analytic continuation in  $\nu$ , and assembles a polynomial  $r_{\text{BFK}}$  by Zagier's formula [3].  $r_{\text{BFK}}$  is finite and well-defined. The issue is that it is a *different* finite polynomial from  $C_S'^{\hat{\Delta}}$ :  $r_{\text{BFK}}$  satisfies the involution  $(1 + S) = 0$  (a direct consequence of BFK's functional equation  $L^*(s) = i^k L^*(k - s)$ ) but *fails* the three-term cocycle relation  $(1 + U + U^2) = 0$  — by  $\sim 22\times$  the typical coefficient magnitude in the numerics. Hence  $r_{\text{BFK}} \notin W$ , while  $C_S'^{\hat{\Delta}} \in W$  by construction.

Structurally, BFK's single-parameter regulator  $e^{iu\tau}$  has  $S$ -symmetry ( $S$  exchanges the two cusps  $\{0, i\infty\}$ , and the regulator at one cusp maps to a regulator at the other) but no analogous symmetry for the  $U$ -orbit  $\{0, 1, i\infty\}$ . Consequently, the cocycle relation forced by  $S^2 = (ST)^3 = 1$  on genuine  $\Gamma$ -cocycles survives the BFK regularisation only in part. The DR finite-endpoint cocycle has no such asymmetry: both contours  $\gamma^{-1}\tau_0 \rightarrow \tau_0$  stay strictly inside  $\mathbb{H}$ , and the cocycle relation holds at the level of  $\Gamma$ -cocycles, not just at the level of the regularised values.

### 6 Verified numerical values

All values below are extracted from the executed notebook at 45-digit working precision and reported here to  $\sim 22$  significant digits.

|                          | Extracted (notebook)       | Brown [1] §8.3            |
|--------------------------|----------------------------|---------------------------|
| $\omega^+(\Delta)$       | -68916772.8095951947543... | -68916772.809595194754... |
| $\omega^-(\Delta)/i$     | -5585015.37931040186688... | -5585015.3793104018668... |
| $\eta^+(\hat{\Delta})$   | 127202100647.177094777...  | 127202100647.17709477...  |
| $\eta^-(\hat{\Delta})/i$ | 10276732343.6491327508...  | 10276732343.649132750...  |

Relative errors vs. Brown's reported values:  $|\Delta\omega^+|/|\omega^+| \approx 4.5 \times 10^{-21}$ ,  $|\Delta\omega^-|/|\omega^-| \approx 1.4 \times 10^{-20}$ ,  $|\Delta\eta^+|/|\eta^+| \approx 8.6 \times 10^{-32}$ ,  $|\Delta\eta^-|/|\eta^-| \approx 9.0 \times 10^{-32}$ .

Basepoint independence (shifting  $\tau_0$  from  $0.3 + 1.2i$  to  $-0.2 + 0.9i$ ) introduces shifts of at most  $2.6 \times 10^{-38}$  for  $\omega^\pm$  and  $1.2 \times 10^{-32}$  for  $\eta^\pm$ , confirming cohomological invariance to full working precision.

## 7 The unified formula

For any weight- $k$  modular form  $f$  on  $\mathrm{SL}_2(\mathbb{Z})$  (holomorphic or weakly holomorphic, regular at interior of  $\mathbb{H}$  or at the cusp), the DR cocycle provides a well-defined period:

$$\omega^\pm(f) = \frac{[C_S^f]_{X^{k-2-j}Y^j}}{[P_S^\pm]_{X^{k-2-j}Y^j}}, \quad \text{any interior monomial } j. \quad (27)$$

For  $f \in S_k$  this equals the classical period  $\int_0^{i\infty} f(\tau)\tau^j d\tau$  times a known algebraic factor. For  $f \in S_k^!$  it is the DR-regularised analogue. In both cases it lies in  $H_{\mathrm{cusp}}^1(\Gamma; V_{k-2})$  and satisfies the full period relations  $(1+S)=0$  and  $(1+U+U^2)=0$ , properties that BFK's single-parameter regularisation cannot guarantee.

## A From cocycles to polynomial-kernel period integrals

This appendix unpacks the cohomological structure behind the construction  $C_S^f = C_S^f - (P|_S - P)$ : what  $P$  actually is, why it has to depend on  $f$ , and how the whole machinery collapses to the cleanest possible statement —  $\omega^\pm(f)$  are finite contour integrals of  $f$  against nine universal  $\mathbb{Q}$ -polynomial kernels. Read as a parallel to the classical fact  $\omega^\pm(\Delta) \in \mathbb{Q} \cdot L(\Delta, s) \cdot (2\pi i)^\bullet$ : in the DR setup, the periods are obtained by integrating  $f$  against fixed polynomial weights on the  $S$ - and  $T$ -segments in  $\mathbb{H}$ , and the lockstep rationality among critical  $L$ -values is encoded as the fact that five different integrals all give  $\omega^-$  and four different integrals all give  $\omega^+$ .

### A.1 Cocycles and coboundaries of $\Gamma$ in $V_{10}$

A 1-cocycle of  $\Gamma$  with values in  $V_{10}$  is a map  $C : \Gamma \rightarrow V_{10}$  satisfying

$$C_{\gamma_1\gamma_2} = C_{\gamma_1}|_{\gamma_2} + C_{\gamma_2}, \quad \forall \gamma_1, \gamma_2 \in \Gamma. \quad (28)$$

Since  $\Gamma = \langle S, T \rangle$  with relations  $S^2 = (ST)^3 = -I$ , a 1-cocycle is determined by the pair  $(C_S, C_T) \in V_{10} \times V_{10}$  subject to the constraints inherited from those relations.

A 1-coboundary is the special cocycle attached to a fixed  $P \in V_{10}$ ,

$$(\delta P)_\gamma := P|_\gamma - P. \quad (29)$$

One checks  $(\delta P)_{\gamma_1\gamma_2} = (\delta P)_{\gamma_1}|_{\gamma_2} + (\delta P)_{\gamma_2}$  directly, so coboundaries are cocycles. Two cocycles differing by a coboundary are *cohomologous*; the quotient  $H^1(\Gamma; V_{10}) := Z^1/B^1$  is the space of cocycles modulo coboundaries. It is finite-dimensional.

The Diamantis–Rolen  $C^f$  from (4), (5) is a genuine cocycle for any  $f$  holomorphic on  $\mathbb{H}$  (or meromorphic with the contour avoiding poles): the proof is a one-liner, write  $\int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\tau_0} = \int_{(\gamma_1\gamma_2)^{-1}\tau_0}^{\gamma_2^{-1}\tau_0} + \int_{\gamma_2^{-1}\tau_0}^{\tau_0}$  and recognise the first piece as  $C_{\gamma_1}^f|_{\gamma_2}$  after  $\tau \rightarrow \gamma_2\sigma$ , using weight-12 modularity of  $f(\tau)d\tau$  under slash. No regularisation, no analytic continuation.

## A.2 Cuspidality and the role of $P$

The cocycle  $C^f$  contains more data than we ultimately want — we want a polynomial in  $V_{10}$ , not a function  $\Gamma \rightarrow V_{10}$ . The strategy is to replace  $C^f$  by a cohomologous *cuspidal* cocycle whose value at  $S$  is the polynomial we extract.

A cocycle is *cuspidal* iff after a coboundary modification  $C^f \rightarrow C^f - \delta P$  we have  $C_T'^f = 0$ ; equivalently,  $P$  satisfies

$$P|_T - P = C_T^f. \quad (\star)$$

The operator  $\delta_T: V_{10} \rightarrow V_{10}$ ,  $P \mapsto P|_T - P$ , has

- $\ker(\delta_T) = \mathbb{C} \cdot Y^{10}$  (the unique  $T$ -fixed monomial, since  $T$  acts as  $X \mapsto X + Y$ ,  $Y \mapsto Y$ ),
- $\text{im}(\delta_T) = \{Q \in V_{10} : [Q]_{X^{10}} = 0\}$ .

So  $(\star)$  is solvable iff  $[C_T^f]_{X^{10}} = 0$ . By periodicity of  $f$ ,  $[C_T^f]_{X^{10}} = (2\pi i)^{11} \int_{\tau_0-1}^{\tau_0} f d\tau = (2\pi i)^{11} a_f(0)$ ; cuspidality is therefore automatic whenever the constant Fourier coefficient  $a_f(0) = 0$ , which it does for both  $\Delta$  and  $\hat{\Delta}$ .

When solvable,  $P$  is unique modulo  $\mathbb{C} \cdot Y^{10}$ . The convention  $p_{10} := 0$  fixes a representative; alternative choices shift  $C_S'^f$  by  $c(X^{10} - Y^{10})$ , which is itself a coboundary in  $V_{10}$  and so does not change the cohomology class. Hence the  $p_{10}$ -ambiguity is invisible to  $\omega^\pm(f)$  — the interior monomials we read off are completely unaffected.

**The matrix system, explicitly.** Writing  $P = \sum_{j=0}^{10} p_j X^{10-j} Y^j$  and expanding  $P|_T(X, Y) = P(X + Y, Y)$  binomially, the coefficient of  $X^{10-m} Y^m$  on the left of  $(\star)$  is  $\sum_{j=0}^{m-1} \binom{10-j}{m-j} p_j$ . So  $(\star)$  becomes the upper-triangular linear system

$$\sum_{j=0}^{m-1} \binom{10-j}{m-j} p_j = [C_T^f]_{X^{10-m} Y^m}, \quad m = 1, 2, \dots, 10, \quad (30)$$

together with the  $m = 0$  identity  $0 = [C_T^f]_{X^{10}}$  (cuspidality). Equivalently, the closed-form Bernoulli operator (15) gives  $P = \delta_T^{-1}(C_T^f)$  in one shot.

**$P$  depends on  $f$ , linearly.** The whole construction is linear in  $f$ : the chain

$$f \xrightarrow{\text{integrate}} \{M_j(f, \tau_0)\}_{j=0}^{10} \xrightarrow{(2\pi i)^{11} (-1)^j \binom{10}{j}} C_T^f \xrightarrow{\delta_T^{-1}} P_f$$

is a composition of linear maps. At a fixed  $\tau_0 = 0.3 + 1.2i$ :

$$\Delta: |C_T^f|_{\max} \sim 10^8, \quad p_0 \approx -1.5 \times 10^4 + 4.9 \times 10^4 i,$$

$$\widehat{\Delta}: |C_T^f|_{\max} \sim 2 \times 10^{14}, \quad p_0 \sim 10^{10} \text{ in modulus.}$$

The six-order-of-magnitude jump is the  $q^{-1}$  Fourier mode of  $\widehat{\Delta}$  amplifying the moments through  $|q_0|^{-1} = e^{2\pi \operatorname{Im}(\tau_0)} \sim 10^3$ . Same recipe ( $\delta_T^{-1}$ ), wholly different  $P$ -polynomials. This is the answer to the natural question “*is the  $(P|_S - P)$  correction universal or form-dependent?*”: it is form-dependent through  $P$ , but universal in the recipe.

### A.3 $\omega^\pm$ as cohomology coordinates

Putting these together: the map  $f \mapsto [C^f] \in H_{\text{cusp}}^1(\Gamma; V_{10} \otimes \mathbb{C})$  is linear in  $f$ . The target is 3-dimensional at weight 12, spanned by Brown’s classes  $[P^+]$ ,  $[P^-]$ , and the Eisenstein direction (which manifests in our explicit coordinates as the  $X^{10} - Y^{10}$  free direction). The map reads off as

$$f \mapsto \omega^+(f)[P^+] + \omega^-(f)[P^-] + c(f)[\text{Eis}].$$

Brown’s basis-coefficient extraction (21) from interior monomials is the explicit recipe for the cuspidal-coordinate pair  $(\omega^+, \omega^-)$ .

### A.4 Periods as polynomial-kernel contour integrals

Now the payoff. Chasing the chain  $f \rightarrow \{M_j, N_j\} \rightarrow P_f \rightarrow C_S^f \rightarrow \omega^\pm$  end-to-end is a composition of  $\mathbb{Q}$ -linear operations; inverting back to integrals against polynomial weights gives the following compact statement.

**Main formula.** For each interior  $j \in \{1, 2, \dots, 9\}$ ,

$$\omega^\pm(f) = \frac{(2\pi i)^{11}}{[P_S^\pm]_{X^{10-j}Y^j}} \left[ \int_{-1/\tau_0}^{\tau_0} f(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} f(\tau) k_T^j(\tau) d\tau \right] \quad (31)$$

where the  $\pm$  sign is determined by the parity of  $j$  (odd  $\rightarrow -$ , even  $\rightarrow +$ ), and  $k_S^j, k_T^j$  are universal  $\mathbb{Q}$ -polynomial kernels of degree  $\leq 10$  in  $\tau$ , depending only on  $j$  — not on  $f$ , not on  $\tau_0$ , not on the  $\pm$  sign. Concretely the  $S$ -kernel is a pure monomial,

$$k_S^j(\tau) = (-1)^j \binom{10}{j} \tau^j, \quad (32)$$

and the  $T$ -kernel is the explicit polynomial recorded in Table 1 below, computed by applying the closed-form Bernoulli formula (15) to extract  $p_j$  and  $p_{10-j}$ , then forming  $k_T^j(\tau) := \sum_r [\beta_{j,r} - (-1)^j \beta_{10-j,r}] \tau^r$ , where  $\beta_{l,r} = [p_l]_{M_r} / (2\pi i)^{11}$  are the rational Bernoulli-operator weights.

**Lockstep rationality, geometrically.** The five rows with  $\operatorname{sgn} = -$  all yield  $\omega^-(f)$ ; the four rows with  $\operatorname{sgn} = +$  all yield  $\omega^+(f)$ . Five different integrals against five different polynomial weights all evaluate to the same number  $\omega^-(f)$ ; four to the same  $\omega^+(f)$ . This is the “lockstep rationality among  $L$ -values” from the main text, expressed not as identities among  $L(\Delta, n+1)$  for various  $n$  but as identities among contour integrals against polynomial weights.

**The  $\tau_0$  catch.** The kernels  $k_S^j, k_T^j$  are universal — they know nothing about  $\tau_0$ ,  $f$ , or even the  $\pm$  sign — but the integration contours  $[-1/\tau_0, \tau_0]$  and  $[\tau_0 - 1, \tau_0]$  do depend on  $\tau_0$ . The  $\tau_0$ -independence of  $\omega^\pm$  is the cohomological invariance of §4; in this language, two different choices of  $\tau_0$  give different contours but the same definite-integral value, because the difference is the coboundary  $h(\tau_0, \tau_1)|_\gamma - h(\tau_0, \tau_1)$  of (26).

| $j$ | sgn | $[P_S^\pm]_{X^{10-j}Y^j}$ | $k_S^j(\tau)$ | $k_T^j(\tau)$                                                                                |
|-----|-----|---------------------------|---------------|----------------------------------------------------------------------------------------------|
| 1   | −   | 4                         | $-10\tau$     | $-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$  |
| 2   | +   | 1                         | $45\tau^2$    | $-9\tau + \frac{45}{2}\tau^2 - 5\tau^3 - 21\tau^5 + 30\tau^7 - \frac{45}{2}\tau^8 + 5\tau^9$ |
| 3   | −   | −25                       | $-120\tau^3$  | $40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$                             |
| 4   | +   | −3                        | $210\tau^4$   | $12\tau - 105\tau^3 + 105\tau^4 + 63\tau^5 - 105\tau^6 + 30\tau^7$                           |
| 5   | −   | 42                        | $-252\tau^5$  | $-42\tau^2 + 210\tau^4 - 252\tau^5 + 84\tau^6$                                               |
| 6   | +   | 3                         | $210\tau^6$   | $-12\tau + 105\tau^3 - 105\tau^4 - 63\tau^5 + 105\tau^6 - 30\tau^7$                          |
| 7   | −   | −25                       | $-120\tau^7$  | $40\tau^2 - 60\tau^3 - 5\tau^4 + 70\tau^6 - 60\tau^7 + 15\tau^8$                             |
| 8   | +   | −1                        | $45\tau^8$    | $9\tau - \frac{45}{2}\tau^2 + 5\tau^3 + 21\tau^5 - 30\tau^7 + \frac{45}{2}\tau^8 - 5\tau^9$  |
| 9   | −   | 4                         | $-10\tau^9$   | $-5\tau + \frac{7}{2}\tau^2 + 5\tau^4 - 7\tau^6 + \frac{15}{2}\tau^8 - 5\tau^9 + \tau^{10}$  |

Table 1: The nine universal polynomial kernels ( $k_S^j, k_T^j$ ) feeding into (31). Computed exactly in [7] via  $\delta_T^{-1}$  in sympy rationals. Note the  $j \leftrightarrow 10 - j$  symmetries: pairs (1, 9) and (3, 7) share their  $k_T$  identically; pairs (2, 8) and (4, 6) share their  $k_T$  up to overall sign (matching the sign flip in  $[P_S^\pm]_j$ );  $j = 5$  is self-dual.

**Comparison with the classical statement.** For  $f = \Delta$ , the limit  $\tau_0 \rightarrow i\infty$  collapses the  $T$ -segment contribution to zero (since  $\Delta$  is cuspidal), and the  $S$ -segment contour stretches to  $\int_0^{i\infty}$ . The single surviving integral  $\int_0^{i\infty} \Delta(\tau) k_S^j(\tau) d\tau$  is a Mellin transform of  $\Delta$ , evaluating by (8) to a multiple of  $L(\Delta, j+1)$  — recovering the classical formulas of (23), (24).

For  $f = \hat{\Delta}$ , the limit is unavailable (both integrals diverge), but at any finite  $\tau_0 \in \mathbb{H}$  each integral is finite and  $\omega^\pm \equiv \eta^\pm$  is well-defined. The contour pair is the DR-regularised analogue of the classical period integral: the *integral* the user wanted.  $\hat{\Delta}$ ’s “periods” really are periods, in this concrete sense.

## A.5 Worked example: explicit polynomials at $\tau_0 = 0.3 + 1.2i$

To make the form-dependence concrete and demonstrate that (31) produces  $\omega^\pm(\Delta)$  and  $\eta^\pm(\hat{\Delta})$  from the same kernels, we tabulate the coboundary correction  $(P|_S - P)(X, Y) \in V_{10} \otimes \mathbb{C}$  — the polynomial subtracted from  $C_S^f$  to produce  $C_S'^f$  — for both forms at the standard basepoint  $\tau_0 = 0.3 + 1.2i$ . All coefficients are mpmath-computed at 30 decimal digits via the Bernoulli operator (15) acting on numerically integrated moments  $\{M_j(f, \tau_0)\}$ , and shown to 4 significant figures. The computation is in [8].

**Period extraction from  $C_S'^f = C_S^f - (P|_S - P)$ .** Subtracting Table 2 from the moments-via- $N_j$  representation of  $C_S^f$  (4), and then dividing the  $C_S'^f$  interior-monomial coefficients by Brown’s basis values:

- $f = \Delta$  (all five  $j \in \{1, 3, 5, 7, 9\}$ , all four  $j \in \{2, 4, 6, 8\}$  giving identical values to  $\sim 10^{-24}$  relative precision):

$$\omega^- = -5\,585\,015.379310402\dots i, \quad \omega^+ = -68\,916\,772.80959519\dots$$

- $f = \hat{\Delta}$  (same nine interior-monomial extractions, all agreeing to  $\sim 10^{-17}$  relative precision):

$$\eta^- = 10\,276\,732\,343.64913\dots i, \quad \eta^+ = 127\,202\,100\,647.1771\dots$$

| $X^{10-j}Y^j$ | $(P_\Delta _S - P_\Delta)$                 | $(P_{\hat{\Delta}} _S - P_{\hat{\Delta}})$       |
|---------------|--------------------------------------------|--------------------------------------------------|
| $X^{10}$      | $1.549 \times 10^4 - 4.866 \times 10^4 i$  | $-5.539 \times 10^{10} - 1.711 \times 10^{11} i$ |
| $X^9Y$        | $-6.008 \times 10^6 + 1.851 \times 10^7 i$ | $3.200 \times 10^{12} + 3.623 \times 10^{12} i$  |
| $X^8Y^2$      | $4.861 \times 10^7 + 2.454 \times 10^7 i$  | $2.150 \times 10^{13} - 9.347 \times 10^{12} i$  |
| $X^7Y^3$      | $6.029 \times 10^7 - 7.995 \times 10^7 i$  | $-1.436 \times 10^{13} - 6.661 \times 10^{13} i$ |
| $X^6Y^4$      | $-9.673 \times 10^7 - 9.984 \times 10^7 i$ | $-1.258 \times 10^{14} + 1.338 \times 10^{13} i$ |
| $X^5Y^5$      | $-1.175 \times 10^8 + 1.004 \times 10^8 i$ | $1.124 \times 10^{13} + 1.546 \times 10^{14} i$  |
| $X^4Y^6$      | $9.673 \times 10^7 + 9.984 \times 10^7 i$  | $1.258 \times 10^{14} - 1.338 \times 10^{13} i$  |
| $X^3Y^7$      | $6.029 \times 10^7 - 7.995 \times 10^7 i$  | $-1.436 \times 10^{13} - 6.661 \times 10^{13} i$ |
| $X^2Y^8$      | $-4.861 \times 10^7 - 2.454 \times 10^7 i$ | $-2.150 \times 10^{13} + 9.347 \times 10^{12} i$ |
| $XY^9$        | $-6.008 \times 10^6 + 1.851 \times 10^7 i$ | $3.200 \times 10^{12} + 3.623 \times 10^{12} i$  |
| $Y^{10}$      | $-1.549 \times 10^4 + 4.866 \times 10^4 i$ | $5.539 \times 10^{10} + 1.711 \times 10^{11} i$  |

Table 2: The coboundary correction  $(P|_S - P) \in V_{10} \otimes \mathbb{C}$  for  $f = \Delta$  and  $f = \hat{\Delta}$  at  $\tau_0 = 0.3 + 1.2i$ . These are the explicit polynomials subtracted from  $C_S^f$  to produce  $C_S'^f$ . The  $\hat{\Delta}$  coefficients are  $\sim 10^6$  times larger than those for  $\Delta$  — the  $q^{-1}$  Fourier mode of  $\hat{\Delta}$  amplifies each moment by  $|q_0|^{-1} = e^{2\pi \text{Im}(\tau_0)} \sim 10^3$ . Same recipe (Bernoulli operator), same matrix system, wholly different output polynomials. Both columns satisfy the anti- $S$ -invariance  $(P|_S - P)|_S = -(P|_S - P)$  at the level of coefficients  $(P|_S - P)_{X^m Y^{10-m}} = (-1)^{m+1} (P|_S - P)_{X^{10-m} Y^m}$ , visible as the symmetry of the table.

**$C_S'^f$  in Zagier's  $W$ -basis.** Zagier's space  $W_{12} \subset V_{10}$  of period polynomials at weight 12 is 3-dimensional, with the standard basis

$$\begin{aligned} P_0(X, Y) &= X^{10} - Y^{10}, \\ P_1(X, Y) &= X^8 Y^2 - 3X^6 Y^4 + 3X^4 Y^6 - X^2 Y^8, \\ P_2(X, Y) &= 4X^9 Y - 25X^7 Y^3 + 42X^5 Y^5 - 25X^3 Y^7 + 4XY^9, \end{aligned}$$

splitting as  $W_{12} = W_{12}^+ \oplus W_{12}^-$  by parity in  $X$ :  $W_{12}^+ = \langle P_0, P_1 \rangle$  (the 2-dimensional even-in- $X$  sector, which includes the Eisenstein direction  $P_0 = X^{10} - Y^{10}$ ) and  $W_{12}^- = \langle P_2 \rangle$  (the 1-dimensional odd-in- $X$  cuspidal sector). Brown's basis is this basis up to one Eisenstein-mixing shift:  $P_S^+ = -\frac{36}{691} P_0 + P_1$  and  $P_S^- = P_2$ .

The cuspidal cocycle value  $C_S'^f$  lies in  $W_{12} \otimes \mathbb{C}$  by construction (§A.2), and so decomposes uniquely as  $C_S'^f = a_0(f)P_0 + a_1(f)P_1 + a_2(f)P_2$ . Reading off monomial coefficients with  $[P_0]_{X^{10}} = 1$ ,  $[P_1]_{X^8 Y^2} = 1$ ,  $[P_2]_{X^9 Y} = 4$ :

$$a_0(f) = [C_S'^f]_{X^{10}}, \quad a_1(f) = [C_S'^f]_{X^8 Y^2} = \omega^+(f), \quad a_2(f) = \frac{1}{4}[C_S'^f]_{X^9 Y} = \omega^-(f).$$

The  $P_1, P_2$  coordinates are the cohomological-invariant periods; the  $P_0$  coordinate is the Eisenstein/convention coordinate (depends on the choice  $p_{10} = 0$ ).

Numerically at  $\tau_0 = 0.3 + 1.2i$ , the two cocycle values are:

$$\begin{aligned} C_S'^\Delta &= (2.752 \times 10^5 - 6.156 \times 10^5 i) P_0 + (-68\,916\,772.80959519 \dots) P_1 \\ &\quad + (-5\,585\,015.379310402 \dots i) P_2, \end{aligned} \tag{33}$$

$$\begin{aligned} C_S'^{\hat{\Delta}} &= (-1.532 \times 10^{11} + 6.163 \times 10^{11} i) P_0 + (127\,202\,100\,647.1771 \dots) P_1 \\ &\quad + (10\,276\,732\,343.64913 \dots i) P_2. \end{aligned} \tag{34}$$

The  $P_1, P_2$  coordinates are exactly  $(\omega^+, \omega^-)$  for  $\Delta$  and  $(\eta^+, \eta^-)$  for  $\widehat{\Delta}$ ; same formula structure, different form  $f$ , different coordinates.

*Why  $(P|_S - P)$  does not have this structure.* The coboundary  $(P|_S - P)$  from Table 2 satisfies  $(P|_S - P)|(1+S) = 0$  (anti- $S$ -invariant; visible as the symmetry  $(P|_S - P)_{X^m Y^{10-m}} = (-1)^{m+1} (P|_S - P)_{X^{10-m} Y^m}$ ), but generically *not*  $(P|_S - P)|(1+U+U^2) = 0$ . So  $(P|_S - P) \notin W_{12}$ : it is the “non- $W$  part” of  $C_S^f$  that gets removed by the modification. Only the result  $C_S'^f$  lives in  $W_{12}$  and admits the 3-coordinate decomposition (33), (34). The Zagier basis projects onto the cohomologically-meaningful  $\omega^\pm, \eta^\pm$  directly.

**The linear-functional view, fully explicit.** Equivalently, the master formula (31), specialised in turn to  $f = \Delta$  and  $f = \widehat{\Delta}$  at any interior monomial  $j \in \{1, 2, \dots, 9\}$ , becomes the pair of fully explicit identities

$$\omega^\pm(\Delta) = \frac{(2\pi i)^{11}}{[P_S^\pm]_{X^{10-j} Y^j}} \left[ \int_{-1/\tau_0}^{\tau_0} \Delta(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \Delta(\tau) k_T^j(\tau) d\tau \right], \quad (35)$$

$$\eta^\pm(\widehat{\Delta}) = \frac{(2\pi i)^{11}}{[P_S^\pm]_{X^{10-j} Y^j}} \left[ \int_{-1/\tau_0}^{\tau_0} \widehat{\Delta}(\tau) k_S^j(\tau) d\tau + \int_{\tau_0-1}^{\tau_0} \widehat{\Delta}(\tau) k_T^j(\tau) d\tau \right], \quad (36)$$

where the  $\pm$  sign is determined by the parity of  $j$  (odd  $\rightarrow -$ , even  $\rightarrow +$ ), the basis value  $[P_S^\pm]_{X^{10-j} Y^j}$  and the kernel pair  $(k_S^j, k_T^j)$  are read off from Table 1, and the basepoint  $\tau_0 \in \mathbb{H}$  is arbitrary (independence of  $\tau_0$  verified in §4).

The two equations (35) and (36) differ only in which modular form  $f$  appears in the integrand: the kernel polynomials are identical and universal. This is the precise sense in which the (quasi-)periods are linear functionals of  $f$ , and it is the direct analog of the classical statement  $\omega^\pm(\Delta) \in \mathbb{Q} \cdot L(\Delta, j+1) \cdot (2\pi i)^{j+1}$  — but now valid uniformly for both holomorphic  $\Delta$  and weakly holomorphic  $\widehat{\Delta}$ , with no analytic continuation, no regularisation, and no integration to the cusp.

## References

- [1] F. Brown, *A class of non-holomorphic modular forms III*, arXiv:1710.07912 (2017).
- [2] N. Diamantis and L. Rolén, *Period polynomials, derivatives of  $L$ -functions, and zeros of polynomials*, arXiv:1707.04814 (2017).
- [3] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special  $L$ -values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [4] `period_extraction_DR_general.ipynb` (executed Jupyter notebook), companion source file. Implements `extract_periods(coeffs_dict, tau0, k)` — the single  $q$ -series-agnostic entry point — and reproduces the  $\omega^\pm, \eta^\pm$  values reported in Section 3, the basepoint-independence check of Section 4, and the  $L$ -value identification in Section 3. Run at 45-digit mpmath precision via `jupyter nbconvert --execute`.
- [5] `verify_bernoulli_formula.py`, companion script. Confirms that the closed-form Bernoulli operator (15) agrees with the upper-triangular recursion (30) on every coefficient  $p_m, m = 0, \dots, 9$ , at relative residual  $\sim 10^{-42}$  at 45-digit working precision. Used to validate (17) and the entire  $\delta_T^{-1}$  closed-form derivation.

- [6] `verify_omega_plus_formula.py`, companion script. Evaluates each of the five (resp. four) equivalent forms of  $\omega^-(\Delta)$  in (23) (resp.  $\omega^+(\Delta)$  in (24)) by computing  $L(\Delta, s)$  via the BFK incomplete-gamma sum and confirms agreement with Brown's reported value to relative error  $< 6 \times 10^{-42}$ . Source of the  $L_{\text{reg}}(\Delta, s)$  table in Section 3.
- [7] `compute_period_kernels.py`, companion script. Computes the nine universal  $\mathbb{Q}$ -polynomial kernel pairs  $(k_S^j, k_T^j)$  of Table 1 exactly via sympy rationals, by applying the  $\delta_T^{-1}$  Bernoulli operator monomial-by-monomial.
- [8] `compute_numerical_polynomials.py`, companion script. Computes the explicit coboundary corrections  $(P|_S - P)$  of Table 2 and the resulting  $C_S'^f$  polynomials (33), (34) for  $f \in \{\Delta, \widehat{\Delta}\}$  at  $\tau_0 = 0.3 + 1.2i$ , working at 30-digit mpmath precision. Verifies period extraction from all nine interior monomials to lockstep agreement.